

HDOC Day-25

Decreasing Patterns using Switch Statement and Nested Loops

Aim

To write a menu-driven C program using a switch statement and nested loops, without arrays, to display different decreasing patterns selected by the user. Each pattern is implemented using a separate parameterless function with no return value.

Algorithm

1. Start the program.
2. Display a menu containing the four decreasing-pattern options and an Exit option.
3. Read the user's menu choice.
4. Use a switch statement to call the corresponding parameterless function.
5. Inside the selected function, input the number of rows n.
6. Use nested loops to print the required decreasing pattern.
7. Return to the menu after displaying the pattern.
8. If the user selects Exit, terminate the program.
9. For an invalid choice, display an appropriate message and show the menu again.
10. Stop.

Test-Case Table

Test Case	Menu Choice	Input n	Expected Output
1	1	4	Decreasing right-angled triangle: 4, 3, 2, 1 asterisks
2	2	4	Rows: 1 2 3 4; 1 2 3; 1 2; 1
3	3	4	Rows: 4 4 4 4; 3 3 3; 2 2; 1
4	4	4	Rows: 4 3 2 1; 3 2 1; 2 1; 1
5	5	-	Program terminates

C Program

```
#include <stdio.h>
```

```
void starPattern(void);
void naturalPattern(void);
void repeatedPattern(void);
void reversePattern(void);
```

```
int main()
{
    int choice;

    do
    {
        printf("\n===== Decreasing Patterns =====\n");
        printf("1. Decreasing right-angled triangle\n");
        printf("2. Decreasing natural-number pattern\n");
        printf("3. Decreasing repeated-number pattern\n");
        printf("4. Reverse-number triangle\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: starPattern(); break;
            case 2: naturalPattern(); break;
            case 3: repeatedPattern(); break;
            case 4: reversePattern(); break;
            case 5: printf("Program ended.\n"); break;
            default: printf("Invalid menu choice.\n");
        }
    } while (choice != 5);

    return 0;
}
```

```
void starPattern(void)
{
    int n, i, j;
    printf("Enter the number of rows (n): ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}
```

```
void naturalPattern(void)
{
    int n, i, j;
    printf("Enter the number of rows (n): ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}
```

```

    }
}

void repeatedPattern(void)
{
    int n, i, j;
    printf("Enter the number of rows (n): ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", i);
        printf("\n");
    }
}

void reversePattern(void)
{
    int n, i, j;
    printf("Enter the number of rows (n): ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--)
    {
        for (j = i; j >= 1; j--)
            printf("%d ", j);
        printf("\n");
    }
}

```

Compilation

```

(kali㉿kali)-[~/day25]
$ gcc day25.c -o day25

```

Execution

```

(kali㉿kali)-[~/day25]
$ ./day25

1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 1
Enter the number of rows: 4
* * * *
* * *
* *
*

```

```
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 2
Enter the number of rows: 4
1 2 3 4
1 2 3
1 2
1
```

```
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 3
Enter the number of rows: 4
4 4 4 4
3 3 3
2 2
1
```

```
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 4
Enter the number of rows: 4
4 3 2 1
3 2 1
2 1
1
```

```
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 6
Invalid choice.
```

```
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: 5
Program ended.
```

```
(kali@kali)-[~/day25]
$
```

Result

The menu-driven C program for displaying the four decreasing patterns using a switch statement and nested loops was prepared and tested with the planned test cases.